GNN概述

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Outline

1. Motivation
2. About GNN
3. History of GNNs
4. Categories
5. Laplacian matrix
6. References

Successful deep learning architectures

Supervised Models

1. Multilayer Perceptrons
2. Convolutional Neural Networks (CNNs)
3. Recurrent Neural Networks (RNNs)

Unsupervised Models

1. Self-Organizing Maps (SOMs)
2. Boltzmann Machines
3. AutoEncoders

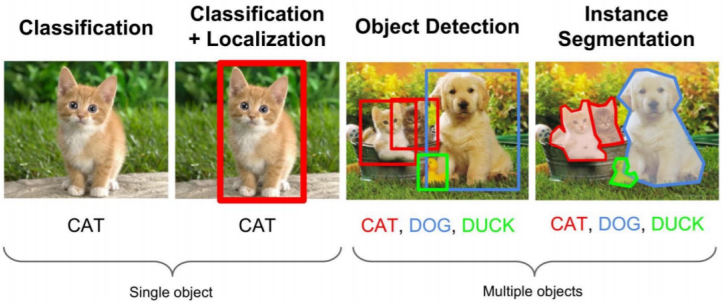
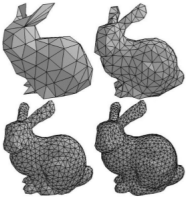


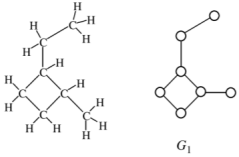
图 1: CNN：深度学习成功应用于各个领域，主要是处理Euclidean空间中图像、文本等结构化数据。

Non-Euclidean data structure (Graph-structured data)

Social Graph (Facebook, Wikipedia)



3D Mesh



Molecular Graph

图 2: All we need is GRAPH! The importance of GNNs is quite significant because there are so many real-world data that can be represented as a graph. Social networks, chemical compounds, maps, transportation systems to name a few (Graph Neural Networks: An overview https://theaisummer.com/Graph_Neural_Networks/).

Networked World

facebook

- 2 billion MAU
- 26.4 billion minutes/day

Alibaba Group 阿里巴巴集团

- >777 million trans. (alipay)
- 200 billion on 11/11

twitter

- 320 million MAU
- Peak: 143K tweets/s

新浪微博 weibo.com

- 462 million users
- influencing our daily life

Instagram

- 700 million MAU
- 95 million pics/day

头条 今日头条

- ~1.5 billion MAU
- 70 minutes/user/day

snapchat

- 300 million MAU
- 30 minutes/user/day

WeChat

- QQ: 860 million MAU
- WeChat: 1.1 billion MAU

图 3: Networked World [ref].

What is GNN

定义 ([1])

Graph neural networks (GNNs) are connectionist models that capture the dependence of graphs via message passing between the nodes of graphs.

定义 (A Gentle Introduction to Graph Neural Networks.)

Graph Neural Network is a type of Neural Network which directly operates on the Graph structure.

定义 (Graph Neural Networks - An overview)

The GNNs are able to model the relationship between the nodes in a graph and produce a numeric representation of it.

General idea of GNN

Notations:

$G = (\mathbf{X}, \mathbf{A})$

1. Feature matrix $\mathbf{X} \in \mathbb{R}^{N \times F}$
2. Adjacent matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$

Input:

1. $\mathbf{X}$
2. $\hat{\mathbf{A}}$

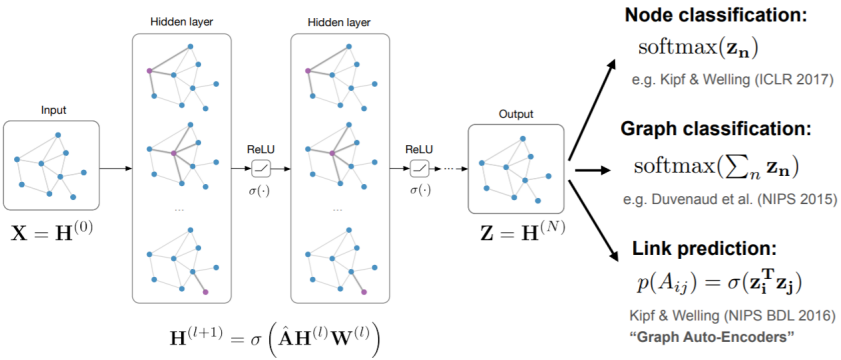


图 4: Main idea: Pass messages between pairs of nodes & agglomerate (ref).

What can we do with GNNs

Classical Problems

1. Node classification
2. Link prediction
3. Node2Vec, Subgraph2Vec, Graph2Vec (Embedding)
4. Learning physics data law

Research Directions

1. Latent graph inference
2. Generative models of graphs

(Potential) Applications more

1. Interacting systems
2. Program induction

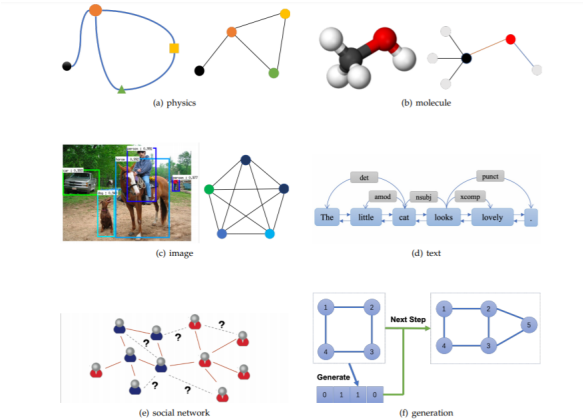


Fig. 4. Application Scenarios

图 5: GNN的应用 [1]

GNN in top conferences

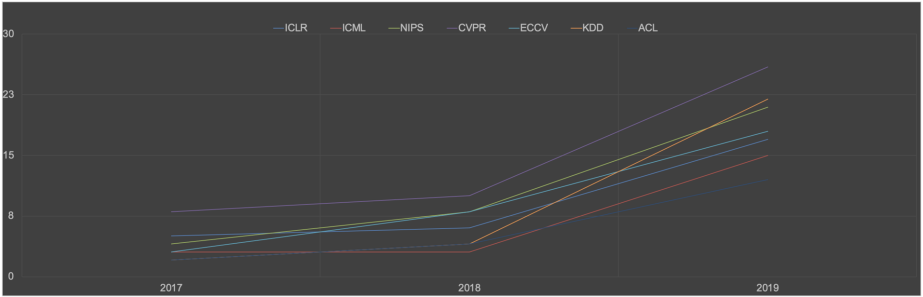
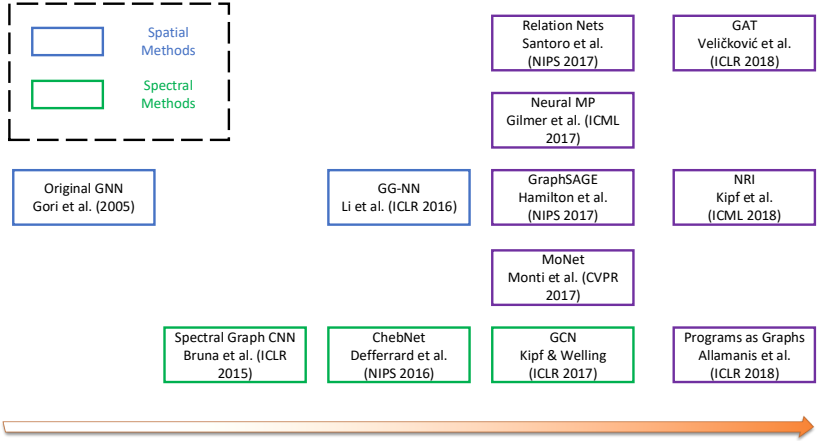


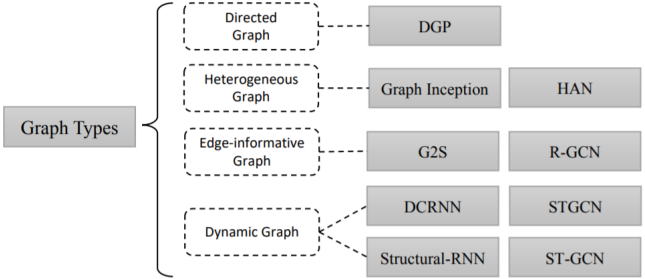
图 6: 各大学术会议纷纷推出 GNN 相关的 workshop，在投中的论文中，以 Graph Network 为关键词的论文数量也呈现井喷之势，上图给出了2017-2019年的情况，上述关键词在各学术会议上的增长趋势。 https://www.infoq.cn/article/LjmbcEgqZV6dzXlFRsjf

据统计，在CVPR 2020中检索到69篇。

A brief history of GNNs



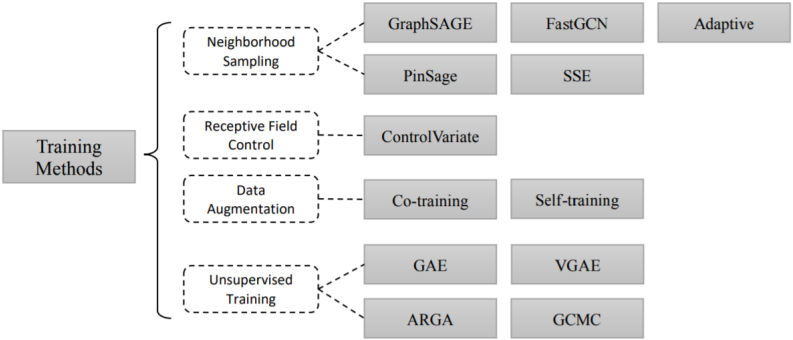
Graph Neural Networks: A Review of Methods and Applications



(a) Graph Types

图 7: 按照Graph Types分类 [1]

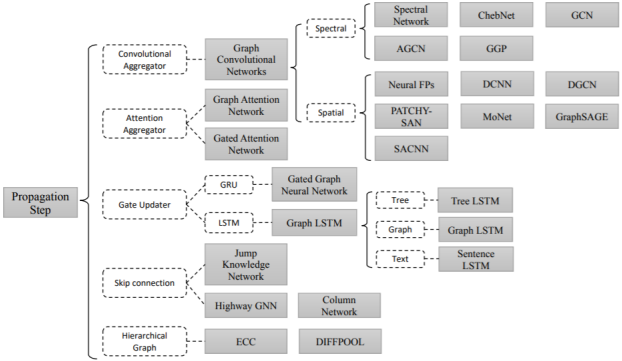
Graph Neural Networks: A Review of Methods and Applications



(b) Training Methods

图 8: 按照Training Methods分类 [1]

Graph Neural Networks: A Review of Methods and Applications



(c) Propagation Steps

图 9: 按照Propagation Steps分类 [1]

A Comprehensive Survey on Graph Neural Networks

这篇综述 [2]将state-of-the-art的GNNs划分为四种：

1. Recurrent Graph Neural Networks
2. Convolutional Graph Neural Networks
3. Graph Autoencoders
4. Spatial-temporal Graph Neural Networks

Highlights:

1. Comprehensive
2. New taxonomy
3. Codes & Datasets

Definition of Laplacian matrix

定义 (Laplacian matrix [a])

[a] Laplacian matrix in WIKIPEDIA: https://en.wikipedia.org/wiki/Laplacian_matrix

给定一个有n个节点的简单图G，它的Laplacian矩阵$\mathbf{L_{n\times n}}$可以定义为

$$\mathbf{L} = \mathbf{D} - \mathbf{A},$$

其中，$\mathbf{D}$是一个degree矩阵，$\mathbf{A}$是一个adjacency矩阵。

$\mathbf{L}$中每一个元素为

$$\mathbf{L}_{i,j} := \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \sim v_j \\ 0 & \text{otherwise} \end{cases},$$

其中，$\sim$表示节点v_i和v_j邻接。

Definition of symmetric normalized Laplacian

定义 (Symmetric normalized Laplacian)

对称归一化Laplacian矩阵被定义为

$$\mathbf{L}^{\text{sys}} := \mathbf{D}^{-\frac{1}{2}}\mathbf{L}\mathbf{D}^{-\frac{1}{2}} = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}.$$

$\mathbf{L}^{\text{sys}}$中每一个元素为

$$\mathbf{L}^{\text{sys}}_{i,j} := \begin{cases} 1 & \text{if } i = j \text{ and } \deg(v_i) \neq 0 \\ -\dfrac{1}{\sqrt{\deg(v_i)\deg(v_j)}} & \text{if } i \neq j \text{ and } v_i \sim v_j \\ 0 & \text{otherwise} \end{cases}.$$

A Short Tutorial on Graph Laplacians, Laplacian Embedding, and ...: https://csustan.csustan.edu/~tom/Clustering/GraphLaplacian-tutorial.pdf

J. Zhou, G. Cui, Z. Zhang, C. Yang, Z. Liu, L. Wang, C. Li, and M. Sun, "Graph neural networks: A review of methods and applications," *arXiv preprint arXiv:1812.08434*, 2018.

Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, "A comprehensive survey on graph neural networks," *IEEE Transactions on Neural Networks and Learning Systems*, 2020.